

ER Diagram

```
erDiagram
    USER {
        int user_id PK
        string username
        string email
        string password
        datetime created_at
    }
    PROJECT {
        int project_id PK
        int user_id FK
        string project_name
        datetime created_at
        datetime updated_at
    }
    FILE {
        int file_id PK
        int project_id FK
        string file_name
        string file_type
        text file_content
        datetime created_at
        datetime updated_at
    }
    VERSION_HISTORY {
        int version_id PK
        int file_id FK
        text previous_content
        datetime saved_at
    }
    USER ||--o{ PROJECT : creates
    PROJECT ||--o{ FILE : contains
    FILE ||--o{ VERSION_HISTORY : stores
```

Entity Description

USER

- Stores user account information.
- A user can create multiple projects.

PROJECT

- Represents a coding workspace.
- Each project can contain multiple code files.

FILE

- Stores project files such as:
 - HTML files
 - CSS files
 - JavaScript files
 - Other programming language files

VERSION_HISTORY

- Stores previous versions of files.
- Helps implement file history and recovery features.

Relationship Summary

```
graph TD; USER -- creates --> PROJECT; PROJECT -- contains --> FILE; FILE -- stores --> VERSION_HISTORY;
```

USER
|
| creates
|
PROJECT
|
| contains
|
FILE
|
| stores
|
VERSION_HISTORY